In-Class 3.4 — Completing the Square

Math 103: College Algebra

Completing the Square

Fill in the blanks to complete the square. Use the squares provided.

1. $x^2 + 8x + \qquad = \left(x + \qquad\right)^2$

x^2	$4x$
$4x$	

2. $x^2 - 4x + \qquad = \left(x + \qquad\right)^2$

x^2	

3. $x^2 - 7x + \qquad = \left(x + \qquad\right)^2$

x^2	

4. $x^2 + \frac{7}{5}x + \qquad = \left(x + \qquad\right)^2$

x^2	

Solve by Completing the Square

Solve each quadratic equation by completing the square.

5. $x^2 - 4x = 8$
6. $x^2 + 3x = 5$
7. $x^2 - 4x + 3 = 0$
8. $x^2 - 7x - 16 = 0$
9. $2x^2 + 8x - 10 = 0$
10. $-\frac{2}{3}x^2 + 6x + 4 = 0$

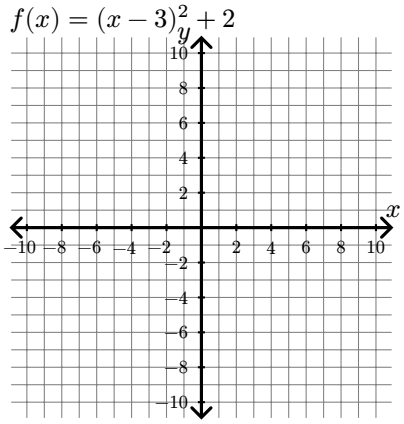
Graphing Parabolas in Vertex Form

For each quadratic function below:

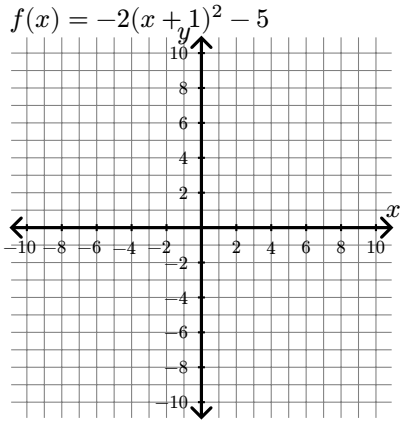
a. Identify and plot the vertex.

b. Plot two symmetric points with integer coordinates on each side of the vertex.

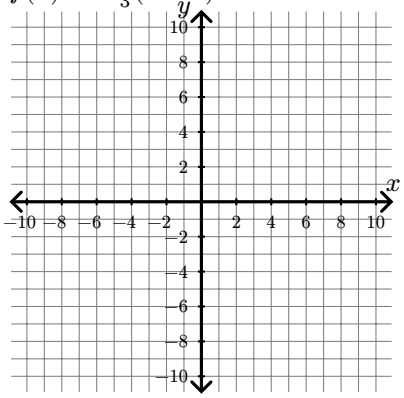
11. $f(x) = (x-3)^2 + 2$



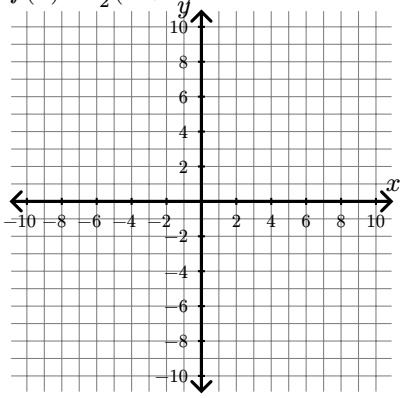
12. $f(x) = -2(x+1)^2 - 5$



13. $f(x) = -\frac{1}{3}(x-2)^2 - 2$



14. $f(x) = \frac{1}{2}(x+5)^2 - 1$



Graphing by Rewriting in Vertex Form

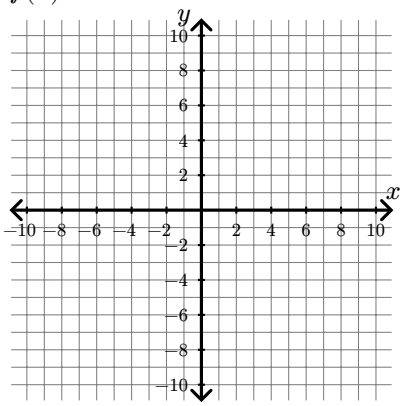
For each quadratic function below:

a. Rewrite the function in vertex form.

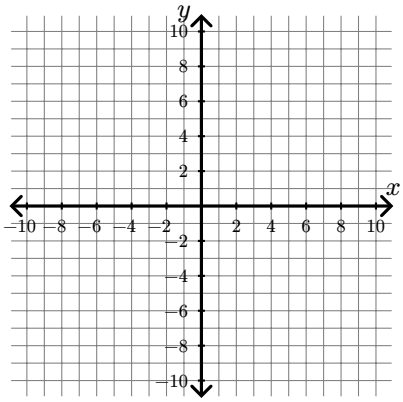
b. Identify and plot the vertex.

c. Plot two symmetric points with integer coordinates on each side of the vertex.

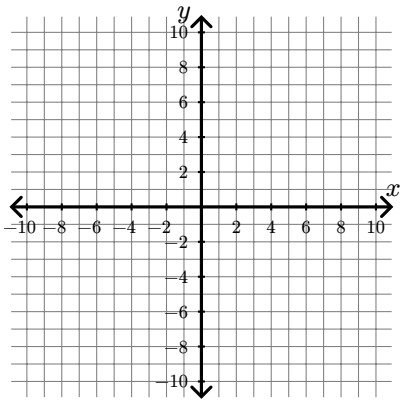
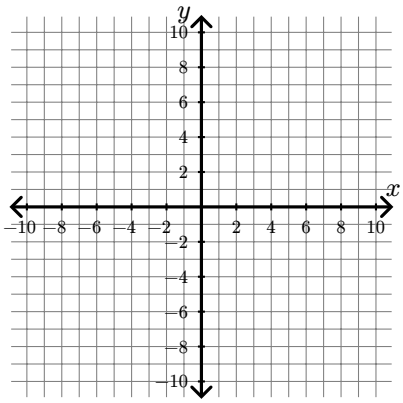
15. $f(x) = x^2 - 6x$



16. $f(x) = -x^2 + 4x + 2$
Hint: $-x^2$ means $-1x^2$



17. $f(x) = 2x^2 + 6x + 5$
Hint: Your points should be $\frac{1}{2}$ and $\frac{3}{2}$ units away from the vertex.



18. $f(x) = -\frac{1}{2}x^2 - 3x - 2$
Hint: Your points should be 1 and 3 units away from the vertex.